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## **Data Acquisition, Modeling and Analysis: Big Data Analytics**

### **HW 8**

#### **Part 2 Observation:**

##### **Scenario 1:**

Changing the rank size had a minimal effect on MSE. The best result recorded was with a rank of 7 that resulted in an MSE of 0.892. Increasing the rank had worsened the MSE score. This suggests that increasing the rank may not always improve the model's performance. The higher rank may have given ALS more latent factors; however, it may have become too large, and the model would then start to overfit on the data.

##### **Scenario 2:**

Increasing maxIter did initially improve the MSE, however once it reached 10 iterations, moving to 20 didn't provide a significant improvement, indicating the performance started to plateau, resulting in further iterations providing minimal gain.

##### **Scenario 3:**

Increasing the data size had the most prevalent effect. The MSE dropped from an MSE of 2.718 with a data size of 2000, to an MSE of 0.8442 with a data size of 100000. This is to be expected, as more training data allows the model to create more patterns from the data.

Overall, the factor that had the greatest effect on MSE was the data size. The second most significant was the number of iterations. Finally, the least significant was the rank size, which initially provided some gain, but then performed worse. It seems ALS benefits the most from having more data, and enough iterations to converge.